

2.4 Sequences and Summations

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We should have seen (or will see)

sequences and summations in Calc II

Definition: A sequence is a function from a subset of the integers (usually $\mathbb{Z}^+$ or $\mathbb{N}$) to a set S . We use a_n to denote the image of the integer n . We call a_n the n^{th} term of the sequence.

Rather than write the sequence as a function we typically write $\{a_n\}$, $\{a_n\}_{n=1}^{\infty}$

so that the i^{th} term of the sequence is a_i .

$$\begin{aligned}\{a_n\}_{n=1}^{\infty} &= \left\{ \frac{1}{n} \right\}_{n=1}^{\infty} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \\ &= \{a_1, a_2, a_3, a_4, \dots\}\end{aligned}$$

Definition: A geometric progression is a sequence of the form

$$\{a_n\}_{n=0}^{\infty} = \{A \cdot r^n\}_{n=0}^{\infty} = \{A, Ar, Ar^2, Ar^3, \dots\}$$

where A is a constant called the initial term

and r is a constant called the common ratio.

$$\{a_n\}_{n=0}^{\infty} = \{(-1)^n\}_{n=0}^{\infty} = \{1, -1, 1, -1, 1, -1, \dots\}$$

$$\begin{aligned}\{b_n\}_{n=0}^{\infty} &= \{2 \cdot (5)^n\}_{n=0}^{\infty} = \{2, 2 \cdot 5, 2 \cdot 25, 2 \cdot 125, \dots\} \\ &= \{2, 10, 50, 250, \dots\}\end{aligned}$$

Definition: An arithmetic progression is a sequence of the form

$$\{a_n\}_{n=0}^{\infty} = \{A + n \cdot d\}_{n=0}^{\infty} = \{A, A+d, A+2d, A+3d, \dots\}$$

where A is a constant called the initial term

and d is a constant called the common difference.

$$\{a_n\}_{n=0}^{\infty} = \{-1 + 4 \cdot n\}_{n=0}^{\infty} = \{-1, -1+4, -1+8, -1+12, \dots\}$$

$$= \{-1, 3, 7, 11, \dots\}$$

$$\{b_n\}_{n=0}^{\infty} = \{7 - 3n\}_{n=0}^{\infty} = \{7, 7-3, 7-6, 7-9, \dots\}$$

$$= \{7, 4, 1, -2, \dots\}$$

Find an expression for the a_n term of the sequences.

$$\{5, -5, 5, -5, \dots\} \quad a_n = (-5)^n \quad n = 0, 1, 2, \dots$$

$$\{1, 4, 7, 10, \dots\} \quad a_n = 1 + 3n \quad n = 0, 1, 2, \dots$$

$$\{-4, 12, -36, 108, \dots\} \quad a_n = 4(-3)^{n-1} \quad n = 0, 1, 2, \dots$$

$$\{-1, 0, 3, 15, 24, 35, \dots\} \quad a_n = n^2 - 1 \quad n = 0, 1, 2, \dots$$

$$\{1, 1, 2, 6, 24, 120, 720, \dots\} \quad a_n = n! \quad n = 0, 1, 2, \dots$$

Definition: A recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of any of $a_0, a_1, \dots, a_{n-1}$. A sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation.

$$\{a_n\}_{n=0}^{\infty} \text{ is defined by } a_n = a_{n-1} + a_{n-2}$$

for $n \geq 2$, $a_1 = 1, a_0 = 0$. → These are the initial conditions of the recurrence relation

$$\{a_n\}_{n=0}^{\infty} = \{0, 1, 1, 2, 3, 5, 8, 13, \dots\}$$

The Fibonacci Sequence!

If we add the terms of a sequence instead of list them we get a sum or summation.

$$\sum_{i=l}^K a_i = a_l + a_{l+1} + \dots + a_{K-1} + a_K$$

$$\sum_{i=l}^K (A \cdot a_i + B \cdot b_i) = A \cdot \left(\sum_{i=l}^K a_i \right) + B \cdot \left(\sum_{i=l}^K b_i \right)$$

EX: Use summation notation to express

the sum of the sequence $\{a_i\}_{i=1}^{100} = \left\{ \frac{1}{i} \right\}_{i=1}^{100}$

$$\sum_{i=1}^{100} \frac{1}{i} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{99} + \frac{1}{100}$$

$$\left| \sum_{i=1}^{100} \frac{1}{i} \right| = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{99} + \frac{1}{100}$$

Ex: Calculate the sum $\sum_{i=1}^4 i^3$.

$$\sum_{i=1}^4 i^3 = 1^3 + 2^3 + 3^3 + 4^3 = 1 + 8 + 27 + 64 = 100$$

Definition: A geometric series is a sum of the form (where A & r are constants)

$$\sum_{i=0}^k A \cdot r^i = A + Ar + Ar^2 + Ar^3 + \dots + A \cdot r^k$$

Theorem: If A & r are non-zero constants

$$\sum_{i=0}^k A \cdot r^i = \begin{cases} \frac{A \cdot r^{k+1} - A}{r-1} & \text{if } r \neq 1 \\ (k+1) \cdot A & \text{if } r = 1 \end{cases} \quad \left(\text{What is } \sum_{i=1}^3 (2(3)^i + 3(2)^i) = ? \right)$$

There are some pretty cool known sums like in the theorem above in Table 2 on page 176 of the book.

What if I have a double sum? Triple sum?

$$\sum_{j=1}^3 \sum_{i=1}^2 (2i-j)$$

Work with a single index at a time

$$\begin{aligned} &= \sum_{j=1}^3 (2-j + 4-j) = \sum_{j=1}^3 (6-2j) \\ &= (6-2) + (6-4) + (6-6) = 4+2 = \boxed{6} \end{aligned}$$